20260521 BEL Semester 2

$\boxed{9}$ Let X_{pijt} denote the quantity of product p shipped from warehouse i to store j on day t, where

$p = 1, \ldots, 2$

$i = 1, \ldots, 3$

$j = 1, \ldots, 4$

$t = 1, \ldots, 7$

a) State the order and dimensions of the tensor X. How many scalar entries does it contain?

$\Rightarrow$ The order (rank) of the tensor X is 4, indexed by the 4 independent variables p, i, j, t.

$\Rightarrow$ The dimensions are $2 \times 3 \times 4 \times 7$!

Total entries: $2 \cdot 3 \cdot 4 \cdot 7 = 168$

b) Define

$$Y_{ij} = \sum_{p=1}^{2} \sum_{t=1}^{7} X_{p,i,j,t}.$$

What are the dimensions of Y, and what does Y_{ij} mean?

$\Rightarrow$ Y_{ij} is a new rank 2 matrix obtained by collapsing (summing) dimension p (products) and dimension t (days).

Dimensions: 3×4

$\Rightarrow$ Y_{ij} means the total cumulative volume of all products combined, shipped from warehouse i to store j across the entire 7-day week.

c) For store $j=2$, define a product demand vector

$$d_p \sum_{i=1}^{3} \sum_{t=1}^{7} X_{pi2t}.$$

What is the dimension of d, and what does each component mean?

Dimension: 2×1

$$d_p = \begin{bmatrix} d_1 \\ d_2 \end{bmatrix}$$

$d_1 \rightarrow$ the total quantity of product 1 delivered to store 2 from all warehouses across all days in the week.

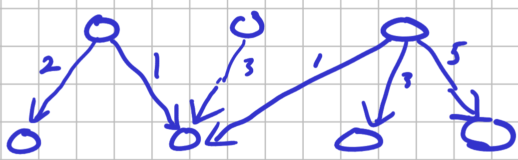
d) For fixed $p = 1$ and $t = 3$, explain what the matrix X_{1ij3} represents.

$\Rightarrow$ This is a 2D slice of the 4-th order tensor. It represents the complete network of warehouse-to-store distribution for product 1 on day 3 of the week.

* This can be seen as a snapshot in time for one product (p_1) in a graph that changes every day.

Day 3 | i (warehouses)

Product 1 | j (stores)



$\boxed{13}$ Every night, stock is redistributed between 2 hubs according to:

$$x_{k+1} = A x_k, \qquad A = \begin{bmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{bmatrix}.$$

Columns represent source hubs, rows represent target hubs.

a) Verify that the columns of A sum to 1. What does this imply operationally?

$\Rightarrow$ Column 1: $0.7 + 0.3 = 1.0$ ✓

Column 2: $0.2 + 0.8 = 1.0$ ✓

What does this imply?

$\Rightarrow$ Because the columns sum to 1, the matrix A is called a column-stochastic matrix.

This implies a total conservation of stock across all nights.

- No accounting for lost/damaged stock
- The total stock between hubs 1 and 2 will be constant from night k to night $k+1$.

b) Compute the characteristic polynomial

$$\det(A - \lambda I)$$

and determine the eigenvalues.

$\Rightarrow$ For matrix A, the characteristic polynomial is

$$\det\left(\begin{bmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right)$$

$$= \det\begin{pmatrix} 0.7-\lambda & 0.2 \\ 0.3 & 0.8-\lambda \end{pmatrix} = (0.7-\lambda)(0.8-\lambda) - (0.3 \times 0.2)$$

$$= 0.56 - 0.7\lambda - 0.8\lambda - \lambda^2 - 0.6 = \lambda^2 - 1.5\lambda + 0.5$$

Eigenvalues are calculated by setting this to zero.

$$\lambda^2 - 1.5\lambda + 0.5 = 0$$

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\Rightarrow \quad \frac{1.5 \pm \sqrt{(1.5)^2 - 4(1)(0.5)}}{2} = \frac{1.5 \pm \sqrt{2.25 - 2}}{2}$$

$$= \frac{1.5 \pm \sqrt{0.25}}{2} = \frac{1.5}{2} \pm \frac{0.5}{2}$$

$$= 0.75 \pm 0.25$$

$$\Rightarrow \quad \lambda_1 = 1 \qquad \lambda_2 = 0.5$$

c) Compute one right eigenvector for each eigenvalue

$\Rightarrow$ We must solv $(A - \lambda I)\vec{v} = \vec{0}$ for each eigenvalue.

λ_1:

$$\begin{bmatrix} 0.7-1 & 0.2 \\ 0.3 & 0.8-1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -0.3 & 0.2 \\ 0.3 & -0.2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

(1) $\quad -0.3v_1 + 0.2v_2 = 0$

(2) $\quad 0.3v_1 - 0.2v_2 = 0$

From (1): $\quad 0.3v_1 = 0.2v_2 \quad \Rightarrow \quad v_1 = \frac{0.2}{0.3} v_2$

$$v_1 = \frac{2}{3} v_2$$

$\Rightarrow$ Choose $v_2 = 3$, that gives $v_1 = 2$

$$\vec{v}_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

λ_2:

$$\begin{bmatrix} 0.7-0.5 & 0.2 \\ 0.3 & 0.8-0.5 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 0.2 & 0.2 \\ 0.3 & 0.3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

(1) $0.2 v_1 + 0.2 v_2 = 0$

$\Rightarrow v_1 = -v_2$

(2) $0.3 v_1 + 0.3 v_2 = 0$

$\Rightarrow$ we choose $v_1 = -1$ $\Rightarrow$ $v_2 = 1$

$$\vec{v}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

d) Interpret the eigenvector corresponding to $\lambda_1 = 1$, and the eigenvector corresponding to the smaller eigenvalue: $\lambda_2 = 0.5$.

$\Rightarrow$ The eigenvector corresponding to λ_1 is

$$\vec{v}_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix},$$

which is the steady-state equilibrium of the system.

Because $\lambda_1 = 1$, any inventory in the ratio 2/3 is preserved over all nights.

$$A\vec{v}_1 = (1)\vec{v}_1.$$

$\Rightarrow$ Eigenvector $\vec{v}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, $\lambda_2 = 0.5$.

This represents a transient imbalance.

Because $\lambda_2 = 0.5$, the imbalance shrinks by half every night, eventually vanishing.

* Note: Decomposing the initial state

A sequence of nights:

$k=0$: $x_{k+1} = Ax_k \Rightarrow x_1 = Ax_0$

$k=1$: $x_2 = Ax_1 = A(Ax_0) = A^2x_0$

$k=2$: $x_3 = Ax_2 = A(A)(Ax_1) = A^3x_0$

$$\boxed{x_k = A^k x_0}$$

We can see x as a vector of vehicles transporting stock between hubs.

The initial state x_0 can be written as a linear combination of our 2 eigenvectors.

$$x_0 = c_1\vec{v}_1 + c_2\vec{v}_2 = c_1\begin{pmatrix}2\\3\end{pmatrix} + c_2\begin{pmatrix}-1\\1\end{pmatrix}$$

$\Rightarrow$ If we let the system run for k nights, we get

$$x_k = A^k x_0 = c_1 A^k\vec{v}_1 + c_2 A^k \vec{v}_2$$

* But we know that $A\vec{v} = \lambda\vec{v}$, so:

$$x_k = c_1(\lambda_1)^k\vec{v}_1 + c_2(\lambda_2)^k\vec{v}_2$$

$$x_k = c_1(1)^k\vec{v}_1 + c_2(0.5)^k\vec{v}_2$$

$$x_k = \underbrace{c_1(1)^k\begin{pmatrix}2\\3\end{pmatrix}}_{\text{Steady}} + \underbrace{c_2(0.5)^k\begin{pmatrix}-1\\1\end{pmatrix}}_{\text{When } k\to\infty \text{ This} \to 0 \text{ Transient!}}$$